

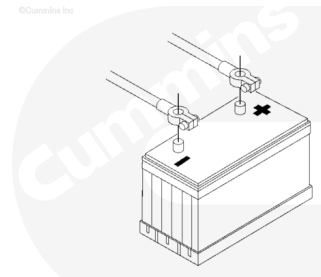
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



13900050

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



ck800wa

⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the cylinder head. Failure to support the engine can cause personal injury.

Note : Use a container that can hold at least 16 liters [27 US qt] of lubricating oil.

- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing
- Remove the starting motor. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html>)
- Remove the transmission, clutch, and all related components (if equipped). Refer to the OEM service manual
- Remove the flywheel/flexplate assembly. Refer to Procedure 016-005

(/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) or Procedure 016-004

(/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)

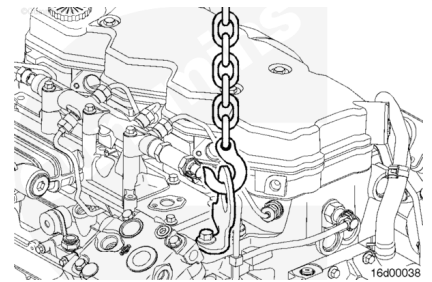
- For rear gear train engines, remove the rear crankshaft seal. Refer to Procedure 001-024
(/qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html)
- For rear gear train engines, disconnect the closed crankcase ventilation hose from the flywheel housing. Refer to Procedure 003-024
(/qs3/pubsys2/xml/en/procedures/40/40-003-024-tr.html)
- Remove any OEM attached components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM instructions.

Remove

All Applications

⚠ WARNING ⚠

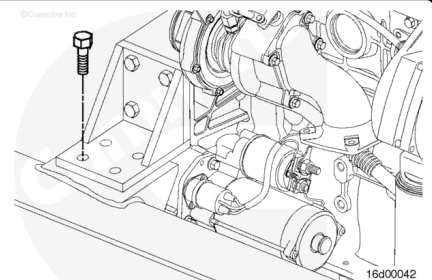
The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



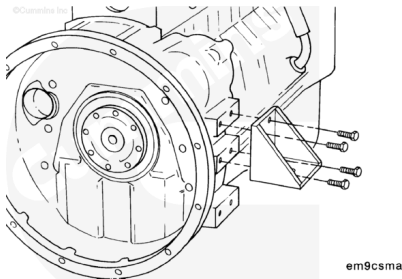
Use a hoist or lifting fixture to support the rear of the engine.

Note : When removing the rear engine mount fasteners, keep track of the location of any shims or spacers used.

Remove the engine mount fasteners.



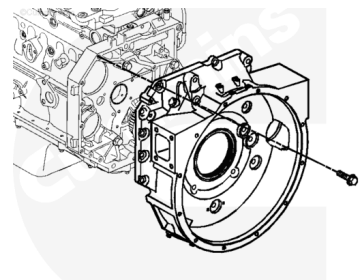
Remove the rear support capscrews and bracket.



Rear Gear Train

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Loosen the flywheel housing capscrews, but do **not** remove.

Using a rubber hammer, loosen the flywheel housing so that the seal is broken between the flywheel housing and rear gear housing.

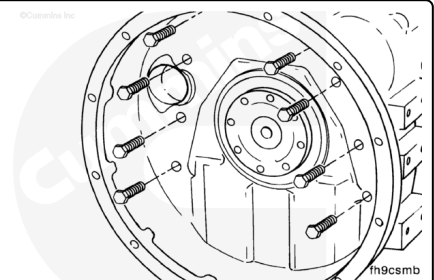
While supporting the flywheel housing, remove the mounting capscrews and the flywheel housing.

Note the location of the flywheel housing capscrews as removed. Some of the capscrews are different length/size fasteners and **must** be installed in the same location as removed.

Front Gear Train

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



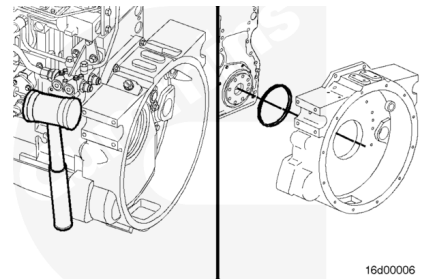
While supporting the flywheel housing, remove the mounting capscrews.

While supporting the flywheel housing, use a rubber hammer to loosen the flywheel housing.

Remove the flywheel housing.

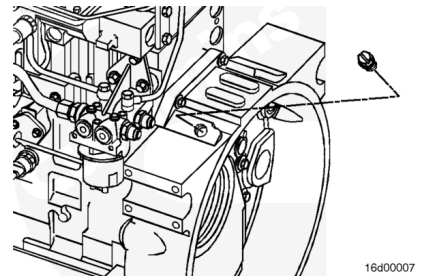
Note : Some engines may have an additional rectangular seal between the flywheel housing and the rear seal carrier

Note : When removing the flywheel housing, note the location of any locating dowel rings.

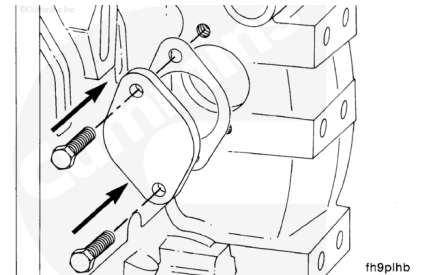


Disassemble

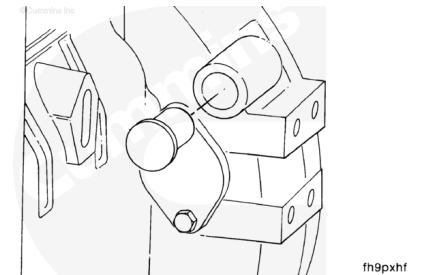
Remove and note the location of any threaded plugs in the flywheel housing.



Remove the access plate and, if equipped, the gasket.



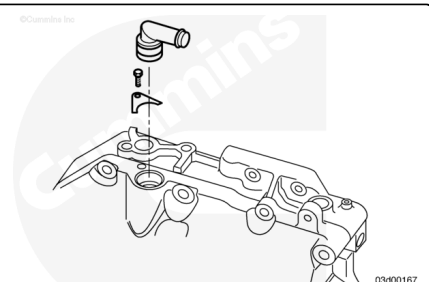
Remove the plug from the barring gear hole.



For rear gear train engines, remove the capscrew securing the closed crankcase ventilation tube adapter clamp plate to the flywheel housing.

Remove the clamp plate.

Remove the closed crankcase ventilation tube adapter.



Clean and Inspect for Reuse

All Applications

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Use only with protective clothing, as well as goggles/shield, and gloves to reduce the possibility of personal injury.

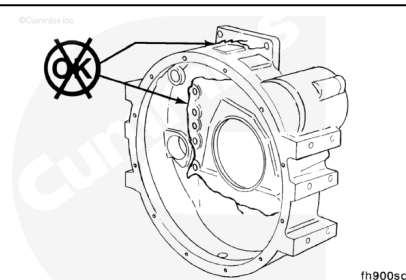
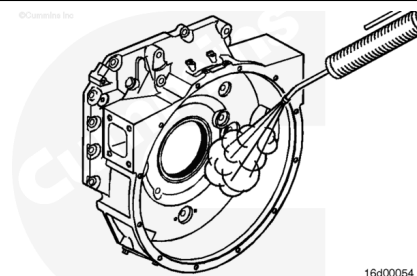
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the flywheel housing.

Dry with compressed air.

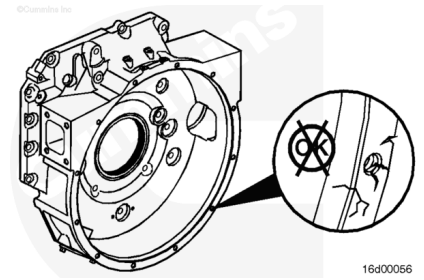
Inspect the flywheel housing for cracks, especially in the area of the flywheel housing that mounts to the cylinder block or rear gear housing.



Inspect the flywheel housing transmission/drive unit mounting surface for cracks.

Also inspect for damaged threads commonly caused by cross-threaded capscrews or installing an incorrect capscrew.

Note : Helicoils are available to repair damaged threads.

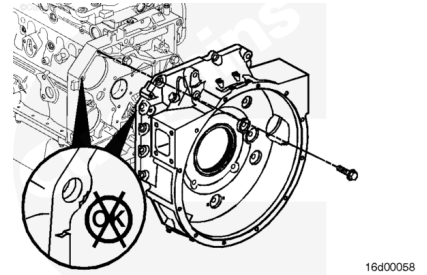


Rear Gear Train

Inspect the rear face of the gear housing and flywheel housing mounting surface for cleanliness and raised nicks or burrs.

Use fine crocus cloth to remove small nicks and burrs.

Thoroughly clean the flywheel housing and gear housing mating surfaces. These surfaces **must** be clean of oil and debris.

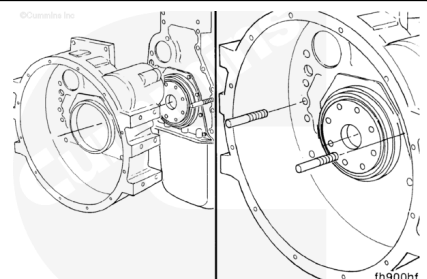


Measure

Note : Follow this step **only** if the flywheel housing (or rear gear housing for rear gear train engines) is being replaced or if troubleshooting a vibration/alignment issue. It is **not** necessary to measure bore alignment or face alignment when installing the original flywheel housing unless the dowel rings were removed during a previous repair.

Install the flywheel housing following the Install Step of this procedure, but do **not** torque the capscrews. Only tighten the capscrews enough to hold the flywheel wheel in place.

Service Tip: For rear gear train engines, when installing a new flywheel housing to check flywheel housing bore alignment and face alignment, do **not** apply sealant to the flywheel housing prior to installing for measurement.



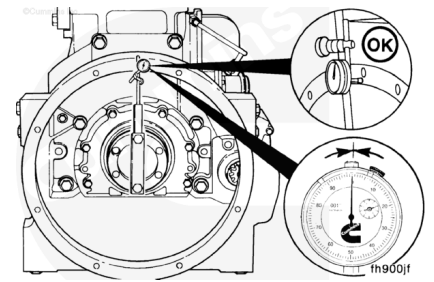
⚠ CAUTION ⚠

When barring the engine using service tool, Part Number 3824591, be careful to not apply excessive side loading to the flywheel housing. This may cause the flywheel housing to move and cause inaccurate measurement readings.

Face Alignment

⚠ CAUTION ⚠

The dial indicator tip must not enter the capscrew holes, or the gauge will be damaged.



Face alignment is determined by calculating the total indicator reading (TIR).

Attach the dial indicator gauge, Part Number 3376050, to the crankshaft. Use mounting tool, Part Number ST1325, to attach the dial indicator to the crankshaft as illustrated.

Note : The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid, so it does not sag. If the bar sags or the indicator slips, the readings obtained will not be accurate.

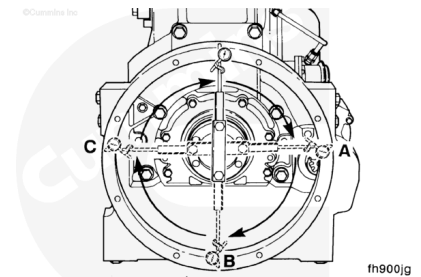
Position the indicator at the 12-o'clock position, and zero the gauge.

Note : The crankshaft must be pushed toward the front of the engine to remove the crankshaft end clearance each time a position is measured.

Using the barring tool, Part Number 3824591, slowly rotate the crankshaft.

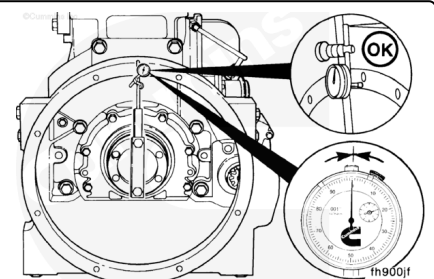
Record the readings at the 3-o'clock (A), 6-o'clock (B), and 9-o'clock positions (C).

The values for A, B, and C could be positive or negative.



Continue to rotate the crankshaft until the indicator is at the 12-o'clock position.

Check the indicator to make sure the needle points to zero. If it does **not**, the readings will be incorrect and the procedure will have to be redone.



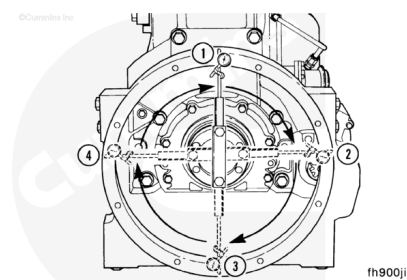
Determine the total indicator reading (TIR).

TIR is determined by calculating the difference between the highest and lowest measurement from the four locations measured.

As the example below illustrates, the TIR would be:

$$+ 0.08 \text{ mm} - (-0.05 \text{ mm}) = 0.13 \text{ mm}$$

$$[+0.003 \text{ in} - (-0.002 \text{ in}) = 0.005 \text{ in}]$$

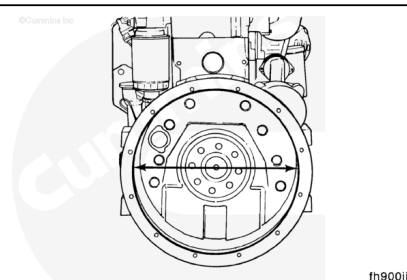


Example:

12 o'clock	0.00 mm	[0.000 in]
3 o'clock	+0.08 mm	[+0.003 in]
6 o'clock	- 0.05 mm	[- 0.002 in]
9 o'clock	+0.08 mm	[+0.003 in]
Equals TIR	0.13 mm	[0.005 in]

The maximum allowable total indicator reading (TIR) is determined by the diameter of the housing bore. If out of specifications, replace the housing.

Note : For rear gear train engines, the rear gear housing may also be the cause of the TIR being out of specification.



Front Gear Train Engine Flywheel Housing Bore and Face Runout

SAE	Bore Diameter		Total Indicator Reading Maximum	
	mm	in	mm	in
00	784.15 to 784.65	30.990 to 31.010	0.48	0.019
0	657.45 to 647.95	25.490 to 25.510	0.41	0.016
1/2	584.00 to 584.40	22.992 to 23.008	0.36	0.014

1	510.98 to 511.38	20.117 to 20.133	0.30	0.012
2	447.55 to 447.81	17.620 to 17.630	0.28	0.011
3	409.45 to 409.71	16.120 to 16.130	0.25	0.010

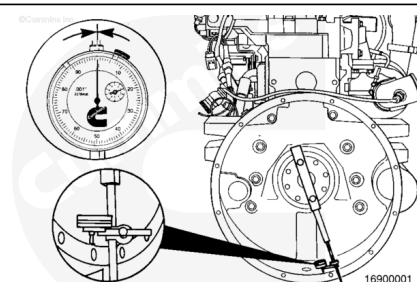
**Rear
Gear
Train
Engine
Flywheel
Housing
Bore and
Face
Runout**

SAE	Bore Diameter		Total Indicator Reading Maximum	
Number	mm	in	mm	in
00	784.15 to 784.65	30.990 to 31.010	0.30	0.012
0	657.45 to 647.95	25.490 to 25.510	0.28	0.011
1/2	584.00 to 584.40	22.992 to 23.008	0.25	0.010
1	510.98 to 511.38	20.117 to 20.133	0.40	0.016
2	447.55 to 447.81	17.620 to 17.630	0.40	0.016
3	409.45 to 409.71	16.120 to 16.130	0.40	0.016

Bore Alignment

Attach the dial indicator gauge, Part Number 3376050, to the crankshaft. Use mounting tool Part Number ST1325 to attach the dial indicator to the crankshaft as illustrated.

Note : The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid, so it does not sag. If the bar sags or the indicator slips, the readings obtained will not be accurate.



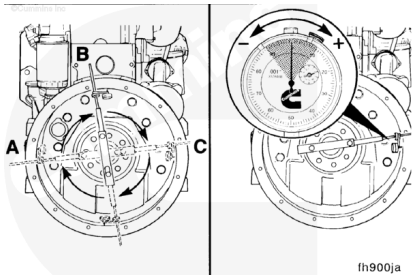
16900001

Position the indicator in the 6-o'clock position, and zero the gauge.

Slowly rotate the crankshaft. Record the readings obtained at the 9-o'clock, 12-o'clock, and 3-o'clock positions as A, B, and C in the concentricity work sheet.

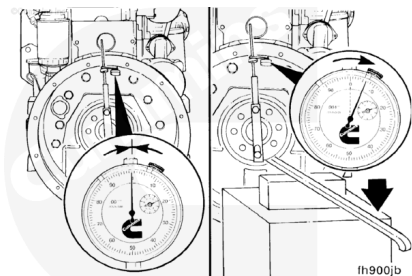
Recheck zero at the 6-o'clock position. If it does **not**, the readings will be incorrect and the procedure will have to be redone.

The values for A, B, and C could be positive or negative. See the accompanying figure to determine the correct sign when recording these values.



⚠ CAUTION ⚠

Do not force the crankshaft beyond the point where the bearing clearance has been removed. Do not pry against the flywheel housing. These actions could cause false bearing clearance readings and result in engine damage.



Rotate the crankshaft until the dial indicator is at the 12-o'clock position and zero the gauge.

Using a pry bar, raise the rear of the crankshaft to its upper limit. Record the value as D on the concentricity work sheet. This is the vertical bearing clearance adjustment, which will **always** be positive.

Create a concentricity work sheet as illustrated to determine the values for the “total vertical” and “total horizontal” values.

Note : The values listed in the concentricity work sheet illustrated are for example only and are listed in inches. The actual numbers measured may differ.

Input the values recorded for A, B, C and D into the concentricity work sheet.

The total horizontal is the 9-o'clock reading, A, minus the 3-o'clock reading, C.

The total vertical is the 12-o'clock reading, B, plus the bearing clearance, D.

Example:

- Six o'clock = reference = 0

Concentricity Worksheet	
9 o'clock	a = 0.004
3 o'clock	c = -0.002
Total Horizontal	a - c = .006
12 o'clock	b = .003
Bearing Clearance	d = .002
Total Vertical	b + d = .005

- Nine o'clock = (a) = 0.004
- Twelve o'clock = (b) = 0.003
- Three o'clock = (c) = (-0.002)

Using the work sheet and the numbers from the example, the total horizontal value equals 0.006 and the total vertical value equals 0.005.

Note : Use the corresponding chart for the SAE 1, 2 or 3 flywheel housings being measured.

Using the illustration, mark the total horizontal value on the horizontal side of the chart and the total vertical on the vertical side of the chart.

Using a straightedge, find the intersection point of the total horizontal and total vertical values. The intersection point **must** fall within the shaded area for the flywheel housing concentricity to be within specification.

Using the total horizontal and total vertical values from the previous example, the intersection point falls within the shaded area. Therefore, the flywheel housing concentricity is within specification.

Note : Make sure to use the correct total indicator reading (TIR) specifications for the flywheel housing being measured when comparing measurements.

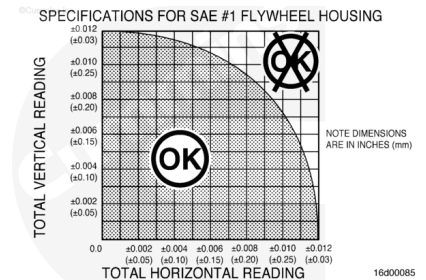


Chart for an SAE 2 flywheel housing.

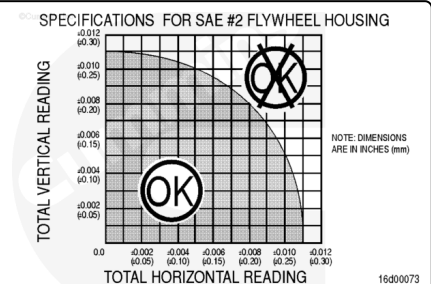
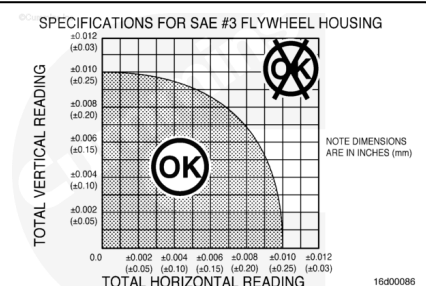
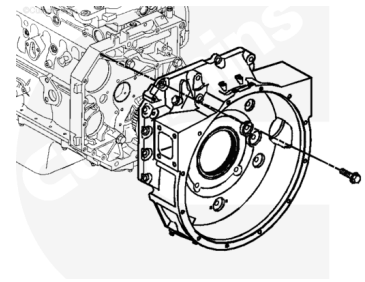


Chart for an SAE 3 flywheel housing.



Do not remove the dowel rings located in the rear gear housing in an attempt to relocate the flywheel housing. The dowel rings are required to locate the rear gear housing to the cylinder block for proper gear alignment. Improper gear alignment will result in engine damage.



16d00053

For rear gear train engine, if the bore alignment is out of specification:

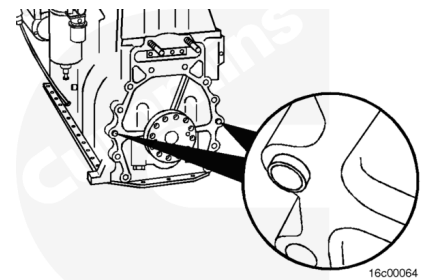
1. Determine if the flywheel housing, rear gear housing or cylinder block has recently been replaced. If any of these components have been replaced, remove and inspect/replace the component.
2. If the flywheel housing, rear gear housing or cylinder block have **not** been recently replaced. Remove the flywheel housing. Inspect the rear gear housing and flywheel housing mounting surfaces. If no damage is found, remove the rear gear housing and inspect the cylinder block and rear gear housing mounting surfaces.

For front gear train engine, if the bore alignment is out of specification:

The ring dowels **must** be removed and the flywheel housing repositioned.

Use the dowel pin extractor service tool, Part Number 3163720, to remove the dowel pins from the block.

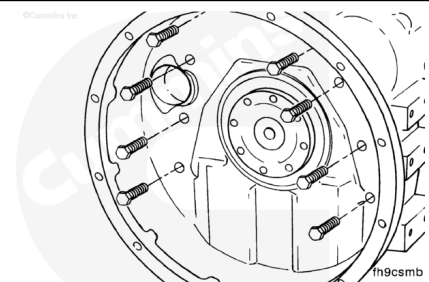
Note : The ring dowels are **not** required to maintain concentricity of the housing; the clamping force of the capscrews holds the housing in place.



16c00064

After the ring dowels are discarded, install the flywheel housing on the engine following the Install Step of this procedure, but do **not** torque the capscrews. Tighten the capscrews enough to hold the flywheel housing in place, but loose enough to allow small movement when struck lightly with a rubber mallet.

Recheck the bore alignment. When bore alignment is within specification, tighten the capscrews to the specified torque value outlined in the Install Step of this procedure.

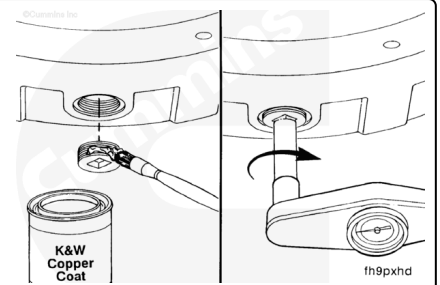


1h9csmb

Assemble

For wet flywheel housings, apply pipe sealant, Part Number 3375066, to any threaded plugs previously removed.

Install and tighten the plugs. Refer to Procedure 017-007 (</qs3/pubsys2/xml/en/procedures/40/40-017-007.html>) for pipe plug torque values for different plug sizes.

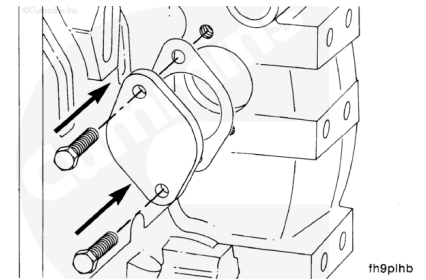


Install the access plate and new gasket.

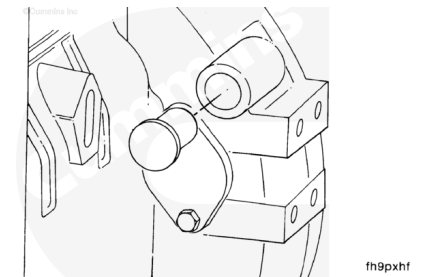
Install the capscrews and tighten.

Torque Value: 24 n•m [18 ft-lb]

Note : If a gasket was **not** previously installed, apply sealant, Part Number 3164067, to the perimeter of the access plate.



With a new o-ring, install the barring gear hole plug.

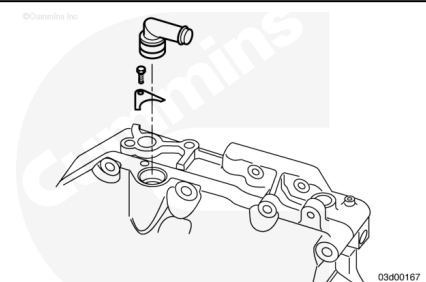


For rear gear train engines, lubricate the o'ring on the closed crankcase ventilation tube adapter with clean engine oil.

Install the breather tube adapter in the flywheel housing.

Install the capscrew and clamp plate to secure the breather tube adapter to the flywheel housing.

Torque Value: 24 n•m [18 ft-lb]



Install

Rear Gear Train

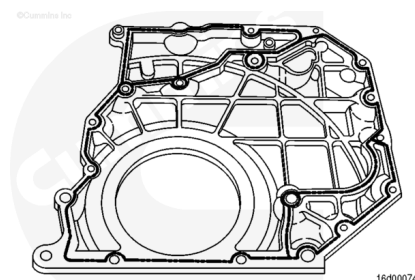
Note : Before installing the flywheel housing, make sure any locating dowel rings are in the same position as when the flywheel housing was removed.

Note : The sealant called for in the following step may appear different than what was originally used to build the engine.

Apply a 1.5 to 2.0 mm [0.06 to 0.08 in] wide bead of sealant, Part Number 3164070, to the back side of the flywheel housing in the path illustrated.

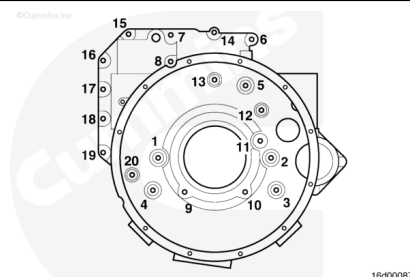
Note : Install the flywheel housing within 10 minutes of applying the sealant or it will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before running the engine.

Install the flywheel housing and capscrews.



Tighten the flywheel housing capscrews in the sequence shown.

Note : Due to different SAE size flywheel housings, the location of the mounting capscrews shown may appear different than the illustration shown. The sequence shown will work for all sizes of flywheel housings.

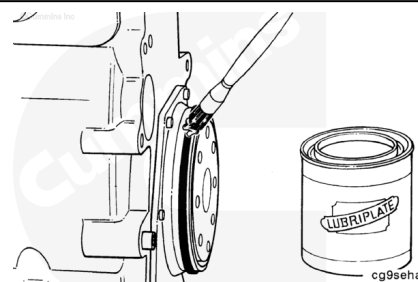


M10	49 n.m	[36 ft-lb]
M12	85 n.m	[63 ft-lb]

Front Gear Train

Note : Before installing the flywheel housing, make sure any locating dowel rings are in the same position as when the flywheel housing was removed.

If previously equipped, install a new rectangular seal on the rear seal carrier and apply assembly lube, Part Number 3163087.

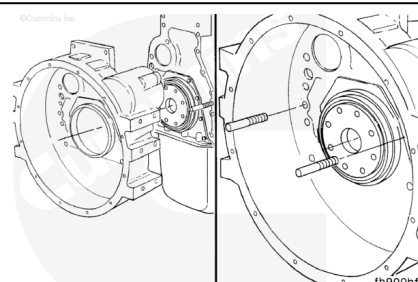


Inspect the rear face of the cylinder block and flywheel housing mounting surface for cleanliness and raised nicks or burrs.

Install two guide pins. Part Number 3163934.

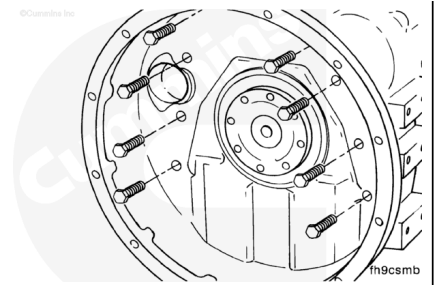
Install the flywheel housing over the guide pins, making sure the flywheel housing is located on the dowel rings.

Note : Be sure the sealing ring is not damaged during installation.



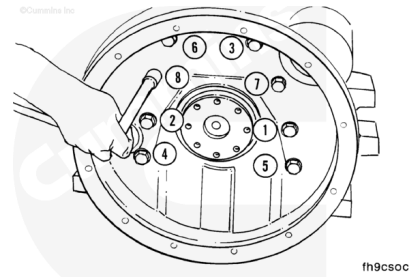
Remove the guide pins.

Install the mounting capscrews.



Tighten the flywheel housing capscrews in the sequence shown.

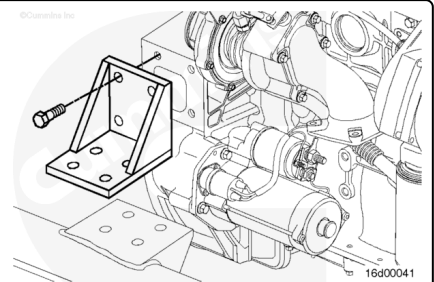
Torque Value: 77 n•m [57 ft-lb]



All Applications

Install the rear engine support bracket and mounting capscrews.

Torque Value: 77 n•m [57 ft-lb]



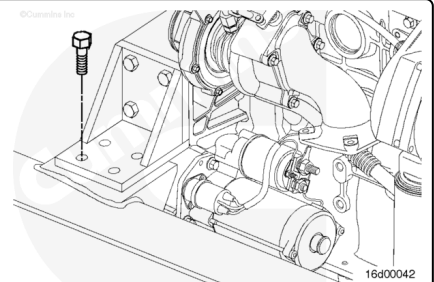
Note : Make sure to install any shims or spacers in the same location as removed.

Lower the rear of the engine.

Install the rear engine mount fasteners.

Tighten to the OEM specification.

Remove the lifting fixture or hoist from the rear of the engine.

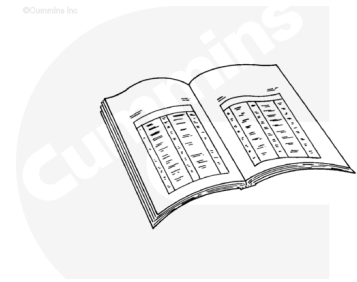


Finishing Steps

- For rear gear train engines, connect the closed crankcase ventilation hose from the flywheel housing. Refer to Procedure 003-024

(/qs3/pubsys2/xml/en/procedures/40/40-003-024-tr.html)

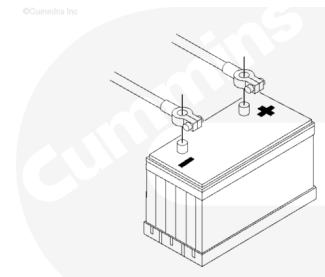
- For rear gear train engines, install the rear crankshaft seal. Refer to Procedure 001-024
(/qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html)
- Install the flywheel/flexplate assembly. Refer to Procedures 016-005
(/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) and Procedure 016-004
(/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the starting motor. Refer to Procedure 013-020
(/qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html)
- Install the transmission and related components (if equipped). Refer to the OEM service manual
- If equipped with a wet flywheel housing, fill the flywheel housing with oil. Refer to the OEM instructions
- If previously removed, attach any OEM components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM instructions.



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Connect the batteries
- Operate the engine and check for leaks.